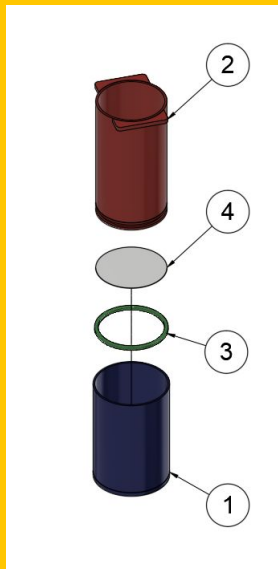


# ME537 Group 1

## A3: Concept

Team Bob the Builders: Will Castro, Jos Moresi, Li Arditi, and Sam Eppley





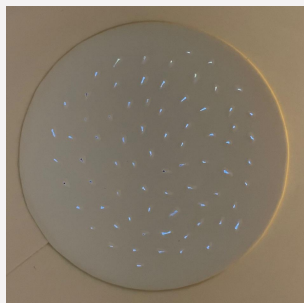
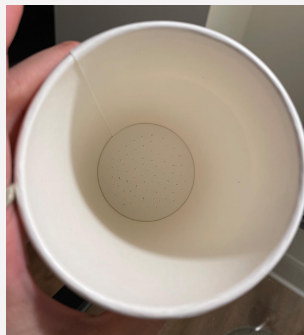
## The Hand Ram

A compact, single serving coffee-brewing device

# Initial Mockups

- Main testing goals:
  - Filtration
  - Manufacturability
  - Reduce Costs

# Initial Mockups



- Lessons Learned
  - Restrict flow along outer wall
  - Shape of cup cannot be tapered (or barrier must account for a taper)
- Next Steps
  - Create cylindrical, concentric walls

# Initial Mockups



- Lessons Learned
  - Fountaining effect upon pressing the plunger
  - Difficult to separate the components
- Next Steps
  - 
  - Add tolerance between components
  - Try a mesh instead of punched holes

# Initial Mockups



- Lessons Learned
  - No fountaining with mesh
  - Easier user experience with a gap between the plunger and vessel
- Next Steps
  - Machining Mockup

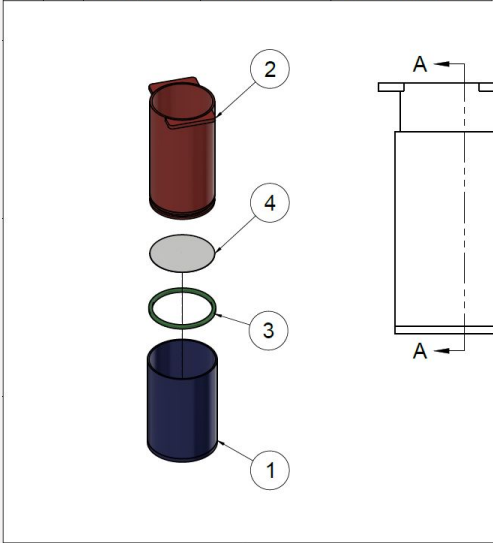
# Initial Mockups



- Lessons Learned
  - Cast at Epic is too short
  - Can't use plastic for cup
  - Must be sealed properly
  - Must be easy to pull out plunger, but enough friction to hold cup & plunger together

# CAD Drawing

| PARTS LIST |     |               |                  |                          |
|------------|-----|---------------|------------------|--------------------------|
| ITEM       | QTY | PART NUMBER   | DESCRIPTION      | MATERIAL                 |
| 1          | 1   | ASMB_VESS-001 | VESSEL ASSEMBLY  | STEEL                    |
| 2          | 1   | ASMB_PLNG-001 | PLUNGER ASSEMBLY | STEEL                    |
| 3          | 1   | SEAL-003_OTS  | X PROFILE SEAL   | RUBBER                   |
| 4          | 1   | FLTR-001_OTS  | FILTER DISC      | STAINLESS STEEL AISI 304 |



| PARTS LIST |     |                    |                |          |
|------------|-----|--------------------|----------------|----------|
| ITEM       | QTY | PART NUMBER        | DESCRIPTION    | MATERIAL |
| 1          | 1   | PLNG_BODY-001_MACH | PLUNGER BODY   | STEEL    |
| 2          | 1   | PLNG_BOT-001_MACH  | PLUNGER BOTTOM | STEEL    |
| 3          | 2   | PLNG_TAB-001_MACH  | PLUNGER TAB    | STEEL    |

